

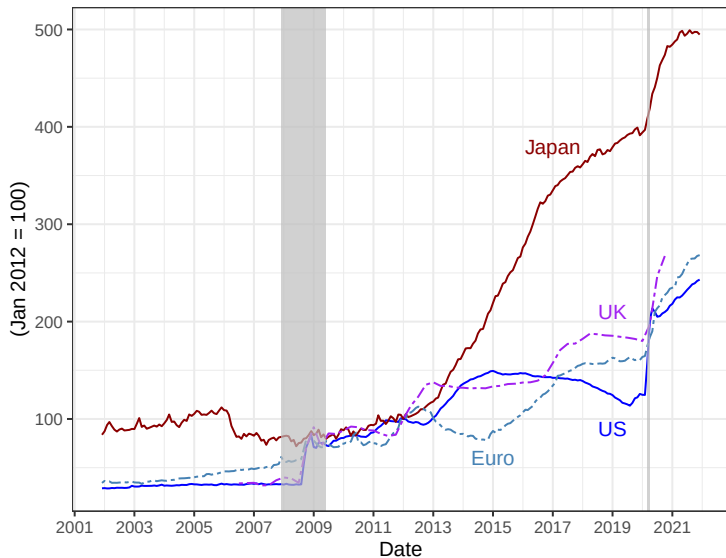
BS Policy Above the ELB by A. Vissing-Jorgensen

Bank of Spain, 2023

by S. Bigio

on July 18, 2025

> Large Balance Sheets



> Motivation

- * What is the optimal size of CB balance sheets?

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- * What is the optimal size of CB balance sheets?
- * Classic answer: Friedman rule
 - * very large balance sheet
 - * satiate banks with reserves
 - * use IOR to control policy
- * V-J paper
 - * estimate reserve/bond demand elasticities
 - * recommend optimal balance sheet size

> Discussion

- * variant of model
- * comment: QE needs seigniorage/fiscal support
- * comment: interference with traditional channel
- * comment: objective function unclear with two assets, costs
- * comments on the empirics:
 - * tighter estimation

Model Discussion

- * Bank Block - Frictionless
- * Exogenous Liquidity Demand
- * Endogenous Liquidity Demand
- * Bonds

> Timing

* two-period: $t = 0, 1$

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- * two-period: $t = 0, 1$
- * focus on static $t = 0$ effects
 - * inflation expectations are anchored
 - * set to meet target

> Notation

- * i nominal rates (between $t = 0$ and $t = 1$)
- * R real rates:

> Notation

- * i nominal rates (between $t = 0$ and $t = 1$)

- * R real rates:

$$R^x = \frac{1 + i^x}{1 + \pi}$$

- * Quantities:

- * lower-case: real

- * upper case: nominal

> Non Banking: Asset Demand System

- * foundations

- * loan demand: working capital loans
- * deposit supply: DIA + quasi-linear good

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Demand System

Deposit supply:

$$d = (R_{t+1}^d)^{\epsilon^d}$$

Loan demand:

$$\ell = \Theta (R_{t+1}^\ell)^{\epsilon^\ell}$$

> Central Bank

- * Standard Instrument (fixed):

$$i^m \rightarrow R^m \equiv \frac{1 + i^m}{1 + \pi}$$

- * Central Bank Balance sheet
 - * private sector loans (Euro)
 - * bonds (US)
 - * reserves
- * Income statement
 - * T^h transfers to households
 - * discount-window loans
 - * purchase loans with M

Model Discussion

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> Bank's Problem | No Frictions

- * Euro model

- * Bank maximizes:

$$\max_{\{\ell, m, d\} \geq 0} \underbrace{R^\ell \ell + R^m m - R^d d}_{\{\text{Portfolio Returns}\}}$$

budget:

$$\ell + m = d$$

> Bank's Problem w/o Frictions

* No frictions | no arbitrage

Return Parity

$$R^\ell = R^m = R^d.$$

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> Bank's Problem | Settlement Frictions

* Portfolio Return:

$$\underbrace{R^\ell \ell + R^m m - R^d d}_{\text{Expected Portfolio Returns}} + \underbrace{\mathbb{E}[\chi(s|\theta)]}_{\text{Expected Settlement Costs}}$$

> Bank's Problem | Settlement Frictions

- * Portfolio Return:

$$\underbrace{R^\ell \ell + R^m m - R^d d}_{\text{Expected Portfolio Returns}} + \underbrace{\mathbb{E}[\chi(s|\theta)]}_{\text{Expected Settlement Costs}}$$

- * Balance at central bank:

$$s = m - \delta d$$

or

$$s = m$$

- * χ : liquidity risk

> χ encodes interbank market

* χ capture settlement costs:

$$\chi(s) = \begin{cases} \chi^- \cdot s & \text{if } s \leq 0 \\ \chi^+ \cdot s & \text{if } s > 0 \end{cases}$$

> Consequences

- * Liquidity service and risk:

$$R^\ell = R^m + \frac{1}{2} \underbrace{[\chi^+ + \chi^-]}_{\mathcal{L}} = R^d + \underbrace{\frac{\delta}{2} \chi^-}_{\mathcal{S}}$$

Liquidity Premia (convenience yield)

$$R^\ell > R^d > R^m$$

- * Exogenous spread $v \equiv \mathcal{L} - \mathcal{S}$
 - * no QE

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* Recall:

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* Tightness (interbank)

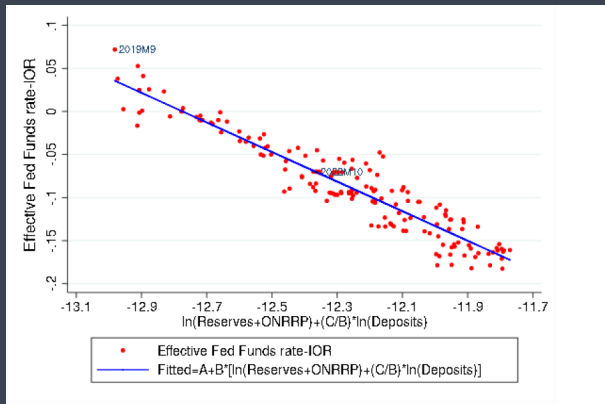
$$\theta = - \frac{\overbrace{m/d - \delta}^{\text{deficit}}}{\underbrace{m/d}_{\text{surplus}}}$$

* $\chi(s; \theta)$ related to tightness

* $\bar{R}(\theta)$ endogenous interbank rate

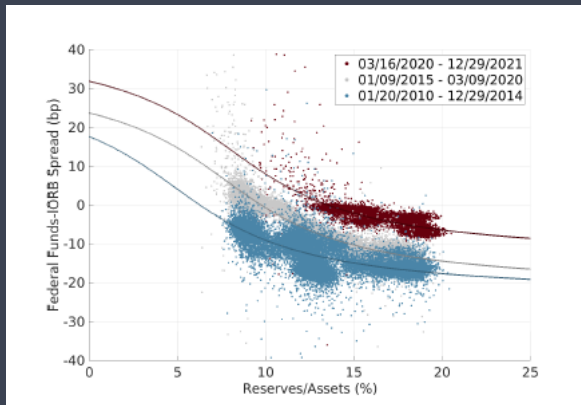
* $\psi^-(\theta)$ discount-window access

> Data Counterparts



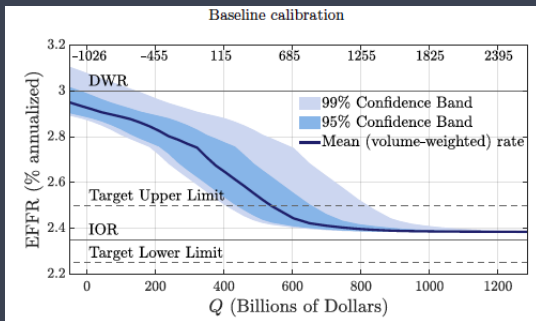
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> Data Counterparts



Afonso, Giannone, La Spada, Williams (2023)

> Data Counterparts



Lagos Navarro (2023)

> Consequences

- * Rates now depend on liquidity service and risk:

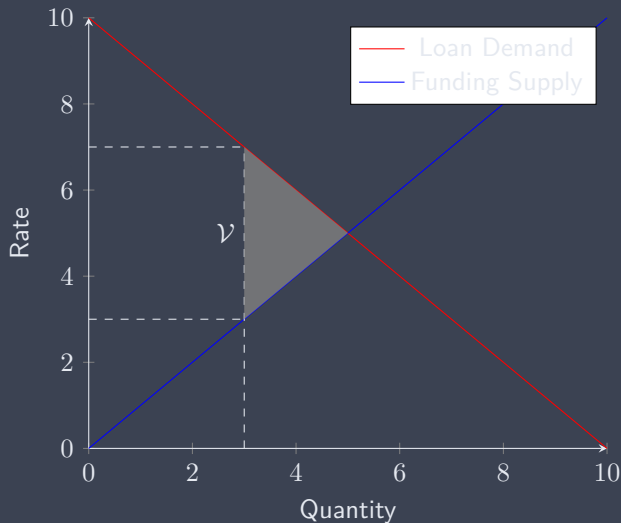
$$R^\ell = R^m + \frac{1}{2} \underbrace{\left[\chi^+(\theta) + \chi^-(\theta) \right]}_{\Sigma(M/P, d)} = R^d + \frac{\delta}{2} \underbrace{\chi^-(\theta)}_{\Sigma(M/P, d)}$$

Recall:

$$\theta = - \frac{\overbrace{M/P/d - \delta}^{\text{deficit}}}{\underbrace{M/P/d}_{\text{surplus}}}$$

$$R^\ell - R^d = \mathcal{V}(M/P, d)$$

> Non Banking: Asset Demand System



> Harberger Triangle

* Why?

$$W(\ell, d) = U(\ell) + V(d)$$

* Then:

$$\Delta \equiv W(\ell^*, d^*) - W(\ell, d) = \int_{\mu_0}^{\mu^*} U'(\ell(\mu)) + V'(d(\mu)) d\mu$$

* Equilibrium conditions:

$$U'(\ell) = R^b - R^m \quad V'(d) = R^d - R^m$$

* Then, we have:

$$\Delta \approx -\frac{1}{2} \left(\underbrace{\mathcal{L}(M/P, d)}_{\text{spread}} (\ell^* - \ell) + \underbrace{\mathcal{S}(M/P, d)}_{\text{spread}} (d^* - d) \right)$$

* Comment: supply/demand elasticities appear in optimal

> Optimal: Flooded Market

* flood interbank market

$$\mathcal{V} = 0$$

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$$\mathcal{V} = 0$$

Friedman Rule

Asset purchase $\mathcal{L}$ under satiation:

$$\uparrow M < \uparrow m(\mathcal{L}) \cdot \underbrace{P}_{\text{fixed}}$$

> Optimal Balance Sheet

- * Comment: need fiscal support (seigniorage or transfers)

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$$\ell = d + e^{cb}(P)$$

- * Return conditions

$$R^\ell = R^m + \mathcal{L}(M/P, d)$$

$$R^d = R^\ell(\ell) + \mathcal{S}(M/P, d)$$

$$d = (R^m + \mathcal{S}(M/P, d))^{\epsilon^d} \rightarrow \bar{d}(M/P)$$

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- * Aggregate resource constraint:

$$\Theta(R^m + \mathcal{L}(m))^{\epsilon^b} = (R^m + \mathcal{S}(m))^{\epsilon^d} + e^{cb}(P)$$

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- * Optimal balance sheet

$$\mathcal{L}(M/P, d) = \mathcal{S}(M/P, d) = 0$$

- * Comment: need fiscal counterpart

> without fiscal counterpart

Neutrality I: Wallace Neutrality

Asset purchase $\mathcal{L}$ under satiation:

$$\uparrow M < \underbrace{m(\mathcal{L})}_{\text{fixed}} \cdot \uparrow P$$

zero-effect on spreads, all to P !

> Loan demand - wage/price rigidity

Price/Wage rigidity

Loan demand (firms):

$$\ell(\bar{R}^\ell, P)$$

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* comment: conflicting targets

> Iso-Fed Funds

- * Iso-Fed Funds:

$$\bar{R} = R^m + \phi(m)$$

- * Then, we

$$0 = dR^m + \phi'(m) \left(\frac{dm}{m} \right)$$

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- * Comment: if target is aggregate demand
 - * right target: R^d

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Demand System

Money Market Funds:

$$f = (R_{t+1}^f)^{e^f}$$

> Shadow Bank's Problem

* Bank maximizes:

$$\max_{\{f, m\} \geq 0} \underbrace{(R^m - R^b) m - (R^f - R^b) f}_{\{\text{Expected Portfolio Returns}\}}$$

$$b + m = f$$

$$b \geq \delta f$$

* We end with:

$$R^m > R^b > R^f$$

> Tradeoffs

- * Much richer responses:

$$b + \Theta \left(R^m + \mathcal{L}(m^b) \right)^{\epsilon^b} = \left(R^m + \mathcal{S}(m^b) \right)^{\epsilon^d} + \delta \overbrace{(b - b^{cb})}^f + e^{cb}(P)$$

- * Bank balance sheet

$$b^{cb} = m + e^{cb}(P)$$

- * Bank reserves

$$m^b = m - \frac{\delta}{1 - \delta} (b - b^{cb})$$

> Harberger Triangle Again

* Welfare now:

$$\Delta \approx -\frac{1}{2} \left(\underbrace{\mathcal{L}(m)}_{\text{spread}} (\ell^* - \ell) + \underbrace{\mathcal{S}(m)}_{\text{spread}} (d^* - d) + \underbrace{\phi(m)}_{\text{spread}} (f^* - f) \right)$$

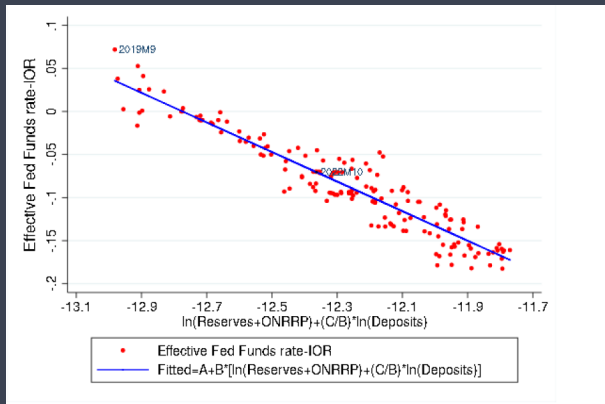
* Increase in $\uparrow m$

$$\uparrow \underbrace{\mathcal{L}(m)}_{\text{spread}} (\ell^* - \ell) + \overbrace{\underbrace{\mathcal{S}(m)}_{\text{spread}} (d^* - d)}^{\text{bank}} + \downarrow \underbrace{\underbrace{\phi(m)}_{\text{spread}} (f^* - f)}^{\text{shadow}}$$

* Comment: here, funding demand elasticities pop up

Empirical Effort

> Data Counterparts



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> Estimation Issues

- * Demand equation:

$$\mathcal{L}(M, D)$$

- * estimates: not H1
 - * problematic for monetary model
- * Potentially polluted by demand shocks:
 - * heterogeneity
 - * demand shifters
- * Use interbank spreads to capture demand

Conclusion

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- * Paper after right question
 - * use money demand elasticities as sufficient statistics

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- * Paper after right question
 - * use money demand elasticities as sufficient statistics
- * Key issues
 - * sufficient stats not enough
 - * demand elasticities are key
 - * conflicting policy goals